Trade Opportunities Report

Data Cleaning & Audit Report — Canada-Brazil (2024-2025)

1. Executive Summary

Metric	Value
Analysis Period	2024-01 to 2025-12
Raw Rows (Import + Export)	89,412
Duplicates Removed	0
Footer Rows Removed	0
Final Rows	89,412
Total Trade Value (CAD)	$27,056,161,445
Export Rows	17,092 ($5,226,473,605)
Import Rows	72,320 ($21,829,687,840)
Suppressed Rows (value=0)	17 — StatsCan confidentiality suppression (see Section 4a)
Chapter 98 Rows (flagged)	182 — kept with is_ch98=True (see Section 4b)
Rows with Zero Quantity	42,034 (47.0%) — confirmed structural, not errors (see Section 5)
UOM Errors	2,247 ($322,460 | 0.001%)
UOM Mismatches	28 ($379,259 | 0.001%)
Value Outliers Flagged	9,373 ($6,266,741,241 | 23.16%)
Final Dataset Columns	28
Quality Checks Passed	18/18 — ALL PASS

2. Raw Data Profile — Before Cleaning

Column structure, data types, and null rate for each source file as received from Statistics Canada. Documents the 'before' state that all subsequent cleaning steps act on.

2a. Import File

File: import-2024-01-2025-12-h10.csv | Rows: 72,320 | Columns: 9

Column	Dtype	Non-Null	Null %
Period	object	72,320	0.0%
Commodity	object	72,320	0.0%
Province	object	72,320	0.0%
Country	object	72,320	0.0%
State	float64	0	100.0%
Value ($)	int64	72,320	0.0%
Quantity	int64	72,320	0.0%
Unit of measure	object	72,320	0.0%
flow	object	72,320	0.0%

2b. Export File

File: export-2024-01-2025-12-h8.csv | Rows: 17,092 | Columns: 9

Column	Dtype	Non-Null	Null %
Period	object	17,092	0.0%
Commodity	object	17,092	0.0%
Province	object	17,092	0.0%
Country	object	17,092	0.0%
State	float64	0	100.0%
Value ($)	int64	17,092	0.0%
Quantity	int64	17,092	0.0%
Unit of measure	object	17,092	0.0%
flow	object	17,092	0.0%

2c. Unit of Measure Distribution — Raw Files

Blank UOM entries are the primary reason quantity cannot be used as a universal metric. Section 5 confirms this is structural, not an error.

Unit of Measure	Export Count	Import Count
Blank	10,744	28,902

Weight in kilograms	4,047	21,968
Number	1,831	13,875
Number of pairs	17	2,108
Weight in metric tonne	291	1,616
Area in square metres	17	1,307
Volume in litres	85	933
Volume in cubic metres	6	523
Number of dozens	3	255
Number in thousands	0	159
Metric tonne air dry	9	149
Weight in grams	6	145
Number of gross	0	135
Volume in litres of pure alcohol	15	109
Length in metres	0	105
Number of packages	0	23
Weight in kilograms of named substance	0	6
Weight in carats	0	2
Radioactivity in megabecquerels	8	0
Radioactivity in gigabecquerels	13	0

3. Cleaning Steps Applied

Every transformation applied to the raw files, in execution order. No rows are removed unless explicitly stated.

#	Step	Detail	Rows / Cols Affected
1	File ingestion	Import + Export CSVs loaded with skiprows=1, skipfooter=3.	89,412 raw rows
2	Footer safety filter	Rows where Period does not match YYYY-MM-DD are dropped. Protects against StatsCan changing footer line count in future exports.	0 removed
3	Column rename	Period, Commodity, Province, Country, Value ($), Quantity, Unit of measure renamed to snake_case.	—
4	State column dropped	Column was 100% null — artifact of StatsCan bilateral-flow template.	1 col removed
5	Deduplication	Exact duplicate rows removed.	0 removed
6	Period parse	period parsed to datetime; year, month, quarter, period_m derived. Original period string excluded from final dataset.	89,412 rows
7	Text cleaning	province, country, uom_text, description: Unicode normalised, non-ASCII stripped [^\x20-\x7E], whitespace collapsed.	4 columns
8	HS hierarchy extraction	hs2/hs4/hs6/hs8/hs10 extracted by digit position from commodity_raw. Stored as zero-padded strings.	89,412 rows
9	HS mapping join	section, section_name, chapter_name joined from hs_mapping.xlsx on hs2.	89,412 rows mapped
10	Negative value assertion	Pipeline aborts if any value or quantity is negative.	0 negatives — PASS
11	Boolean flags	is_ch98 = (hs2=='98') | has_quantity = (uom_text!='Blank') | is_suppressed = (value==0).	ch98=182 | suppressed=17 | no_qty=39646
12	UOM audit	uom_code mapped from uom_text. uom_expected joined from t2026_uom.xlsx. uom_status = OK / Error / Mismatch.	OK=87,137 | Err=2247 | Mis=28
13	Value outlier detection	IQR x1.5 on trade VALUE per hs8 (min 4 obs). lower_bound clipped at 0. Outliers flagged, NOT removed.	9,373 flagged
14	Final column selection	Dataset reduced to 28 columns. Internal columns (uom_expected, _flag_multi_uom, raw period) excluded.	28 cols

4. Data Quality Proof

Automated checks run after all cleaning steps and before the CSV is written. A failing check aborts the pipeline. All expected values are derived from the dataset itself — no hardcoded thresholds.

#	Check	Expected	Result
1	No duplicate rows	0 duplicates	OK
2	No negative values	All values >= 0	OK
3	No negative quantities	All quantities >= 0	OK
4	All 24 periods present	24 unique periods	OK
5	Period range: 2024-01 to 2025-12	Min=2024-01, Max=2025-12	OK
6	flow column has expected values only	Only Export, Import	OK
7	All periods match YYYY-MM format	YYYY-MM for all rows	OK
8	No footer text in period column	No footer text in data	OK
9	All hs2 codes mapped to chapter_name	No NaN in chapter_name	OK
10	hs2 always exactly 2 characters	Exactly 2 characters	OK
11	hs8 always exactly 8 characters	Exactly 8 characters	OK
12	is_suppressed consistent with value==0	value==0 for all True	OK
13	is_ch98 consistent with hs2==98	hs2==98 for all True	OK
14	has_quantity consistent with uom_text	Matches uom_text!='Blank'	OK
15	uom_status has no nulls	No NaN in uom_status	OK
16	uom_code has no UNKNOWN values	No UNKNOWN in uom_code	OK
17	lower_bound_hs8 never negative	>= 0 for all non-null	OK
18	is_outlier never set on suppressed rows	0 overlap	OK

4a. Suppressed Rows — value = 0

Statistics Canada withholds value and quantity when a single company dominates a trade cell in a given month (Disclosure Control Policy). These 17 rows are retained with is_suppressed=True and must be excluded from value aggregations.

Period	Flow	Province	Chapter	Description
2024-07	Export	Quebec	Plastics and Articles	Fittings for furniture, coachwork or the like, of plast
2024-12	Export	Quebec	Printed Books and News	Plans and drawings for architectural etc originals draw
2025-07	Export	Quebec	Electrical Machinery a	Electric conductors, insulated, for a voltage not excee
2024-01	Export	Quebec	Paper and Paperboard	Paper labels of all kinds, printed
2025-06	Export	Ontario	Electrical Machinery a	Light-emitting diode (LED) lamps, electric, o/t modules
2024-08	Export	Ontario	Electrical Machinery a	Insulating fittings of plastics for elec machines, appl

2025-04	Export	British Columbia	Paper and Paperboard	Paper, hand-made, uncoated, in rolls or sheets
2024-09	Export	Quebec	Electrical Machinery a	Electric conductors, insulated, for a voltage exceeding
2024-03	Export	British Columbia	Cereals	Rye, seed for sowing
2025-07	Export	Quebec	Electrical Machinery a	Electro-magnets, nes and parts of No 85.05
2025-09	Export	British Columbia	Articles of Iron and S	Screws and bolts, whether/not with their nuts or washer
2025-07	Export	Quebec	Articles of Iron and S	Washers, spring or lock, iron or steel
2024-06	Export	Quebec	Plastics and Articles	Fittings for furniture, coachwork or the like, of plast
2025-12	Export	Ontario	Inorganic Chemicals	Natural uranium and its compounds, nes
2025-03	Export	Ontario	Electrical Machinery a	Electric conductors, insulated, for a voltage not excee
2025-11	Export	Quebec	Electrical Machinery a	Winding wire, insulated, of copper, electric, including
2024-12	Export	Ontario	Printed Books and News	Plans and drawings for architectural etc originals draw

4b. Chapter 98 — Special Classification Provisions

Chapter 98 is a Canada-specific HS chapter reserved for movements that cross the border outside normal commercial trade. It covers: **Repairs** (goods sent abroad and returned after repair); **Settlers effects** (personal/household goods of immigrants); **Traveller goods** (items acquired abroad by Canadian residents); **Low-value postal shipments** (below the de minimis threshold); **Charitable donations**; and **Returning Canadian-origin goods** (previously exported, reimported unchanged). All 182 rows are retained with is_ch98=True. They must be excluded from commodity trend analysis and forecast models.

Description	Rows	Value (CAD $)
Repairs	44	$85,187,916
Settlers/migrants' personal and household effects, valued at $2,000 CD	30	$937,528
Non Canadian mfd goods of sections V to X, returning w/o any process o	1	$137,646
Non Canadian mfd goods of sections I to IV, returning w/o any process	2	$127,592
Non Canadian mfd goods of section XVII, returning w/o any process of m	1	$20,000
Donation of clothes/books for charity and photographs,3 maximum,not fo	27	$15,453
All goods valued at $2,000 CDN/more, acquired by all categories of tra	1	$5,000
Low value shipment < $500, imported by mail, <= 5% of M.F.N. Tariff	27	$2,687
Low value shipment < $500, imported by mail, within 5% of M.F.N. Tarif	20	$1,285
Non Canadian mfd goods of section XVI, returning w/o any process of mf	1	$1,250
Donations,o/t tobacco or alcohol,sent/brought in by non resident,<=$60	14	$1,156
Low value shipment < $500, imported by mail, within 5% of M.F.N. Tarif	5	$642
Non Canadian mfd goods of sections XI to XII, returning w/o any proces	1	$391
Non Canadian mfd goods of sections XIII to XV, returning w/o any proce	1	$27
Goods owned and used before settler's arrival,accompanying when enteri	6	$25

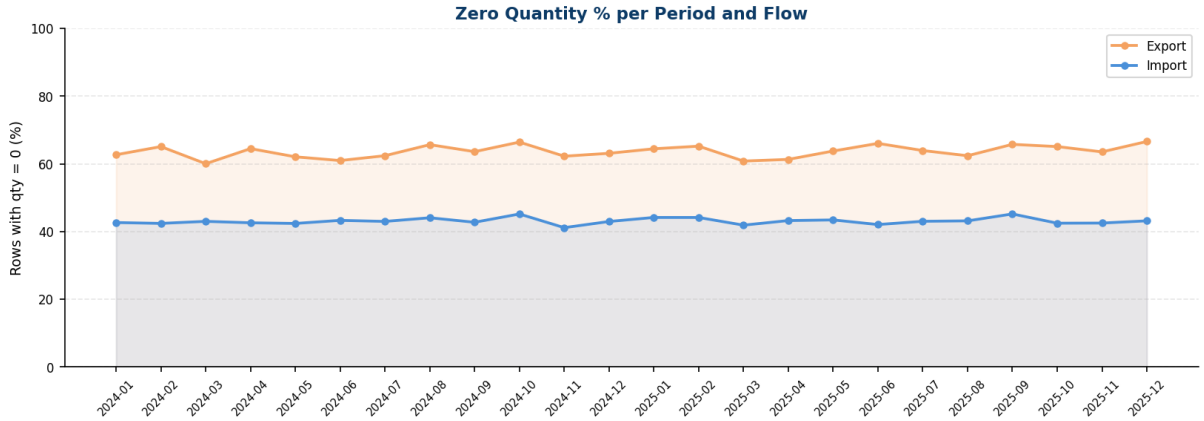
Goods of seasonal residence,imported by person who is not a CDN reside	1	$3

5. Primary Metric Decision — Value vs Quantity

During data profiling, 47.0% of rows had quantity=0. Before selecting the primary analysis metric, we investigated whether these zeros were data errors or a structural characteristic of the source.

Finding 1 — Zero-quantity % is stable across all 24 months

If zeros were caused by data errors or processing failures, their proportion would vary irregularly across months. The chart below shows the opposite: the rate is stable for every flow.



Export: avg 63.7% zero-qty/month (std 1.9%, range 60.1–66.6%) | Import: avg 43.1% zero-qty/month (std 1.0%, range 41.2–45.2%). Low std dev confirms the pattern is structural.

Finding 2 — The zeros are caused by StatsCan Blank UOM

The majority of zero-quantity rows have uom_text='Blank' — StatsCan's explicit code indicating no physical unit is published for that commodity. This is a regulatory data collection decision, not a missing value.

Cause	Rows	% of zero-qty rows
uom_text = Blank — StatsCan does not publish quantity for this commodity	39,646	94.3%
uom_text defined but quantity = 0 — rounding or low-volume suppression	2,388	5.7%
value = 0 AND quantity = 0 — full confidentiality suppression	17	0.0%

Decision and consequence

Decision: Trade VALUE (CAD $) is the primary metric for all analyses. Quantity is retained in the dataset for reference but is not used in aggregations. The has_quantity flag allows filtering to rows where quantity is available if needed for future volume analyses.

Consequence for outlier detection: A unit-price approach (value / quantity) would produce NaN for 47.0% of rows where qty=0, giving incomplete coverage. This pipeline applies IQR outlier detection directly on VALUE per hs8, giving 100% coverage over all non-suppressed rows.

6. Trade Value Statistics by HS Chapter

Transaction-level VALUE (CAD $) statistics grouped by HS Chapter (hs2). Suppressed rows (value=0) excluded. Sorted by total chapter value descending. Value distribution is right-skewed (positive skew) — a small number of high-value transactions dominate. Kurtosis > 3 indicates heavy tails (extreme single-transaction values).

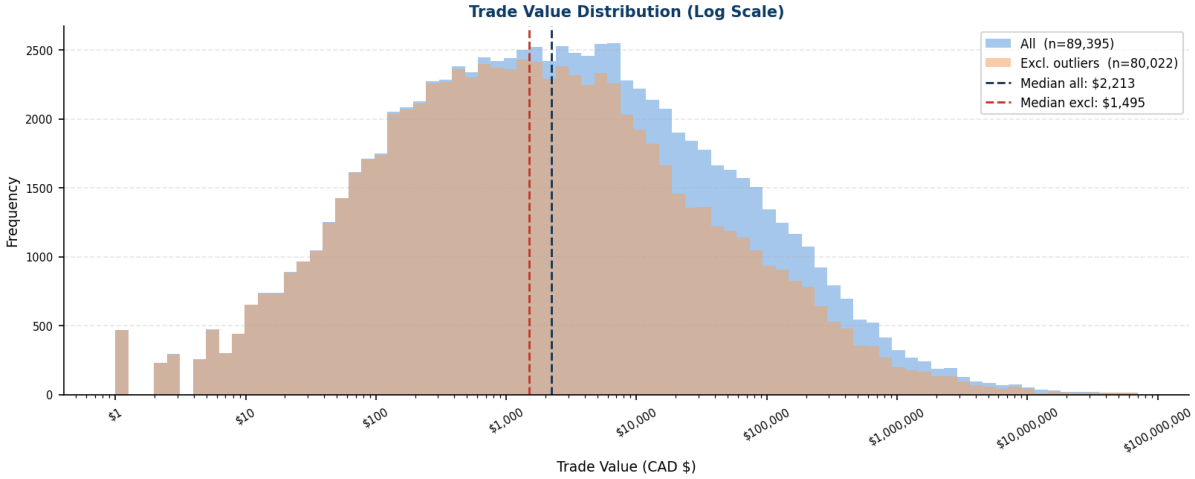
HS2	Chapter	n	Total (CAD $M)	Min	Median	Mean	Max	Skew	Kurt
71	Precious Stones, Metals and	797	$7,005.7M	$4	$2,797	$8,790,041	$570,887,584	+7.3	+64.5
28	Inorganic Chemicals	790	$4,580.8M	$1	$27,841	$5,798,523	$373,371,415	+6.6	+48.0
31	Fertilizers	130	$2,725.7M	$3	$119,144	$20,966,571	$218,192,119	+2.4	+4.9
84	Nuclear Reactors and Machine	16,477	$2,283.9M	$1	$3,527	$138,613	$41,319,265	+19.3	+513.3
17	Sugars and Sugar Confectione	745	$1,811.9M	$1	$8,149	$2,432,089	$119,335,340	+5.8	+40.4
88	Aircraft and Spacecraft	360	$1,291.4M	$88	$68,229	$3,587,219	$132,122,738	+5.7	+35.0
09	Coffee, Tea, Mate and Spices	1,062	$877.6M	$1	$3,186	$826,334	$34,115,132	+5.6	+34.9
72	Iron and Steel	1,317	$665.0M	$5	$10,549	$504,912	$61,328,737	+12.4	+169.2
47	Pulp of Wood and Cellulosic	160	$558.9M	$6	$2,278,006	$3,493,279	$14,068,248	+0.8	-0.5
85	Electrical Machinery and Equ	9,216	$553.5M	$1	$1,864	$60,056	$12,247,831	+17.0	+445.5
20	Vegetable and Fruit Preparat	1,371	$518.9M	$1	$5,519	$378,477	$13,342,357	+5.6	+34.6
26	Ores, Slag and Ash	140	$447.3M	$1	$126	$3,194,828	$62,790,671	+4.3	+24.8
30	Pharmaceutical Products	1,006	$314.0M	$1	$200	$312,099	$48,629,167	+19.5	+490.9
02	Meat & Edible Offal	379	$276.5M	$146	$270,200	$729,661	$6,602,126	+2.5	+6.9
90	Optical, Medical and Precisi	4,402	$192.9M	$1	$3,575	$43,828	$5,167,469	+14.3	+375.4
40	Rubber and Articles Thereof	3,630	$181.7M	$1	$1,135	$50,059	$3,809,732	+8.6	+100.0
27	Mineral Fuels and Oils	143	$176.9M	$1	$23,644	$1,237,170	$18,193,308	+3.0	+9.4
87	Motor Vehicles and Parts	3,881	$167.7M	$1	$1,450	$43,221	$4,970,660	+10.5	+152.5
25	Salt, Sulfur, Stone and Ceme	518	$149.6M	$1	$5,036	$288,745	$27,871,140	+12.6	+195.6
08	Edible Fruit and Nuts	923	$149.3M	$1	$28,569	$161,800	$6,286,906	+7.5	+68.2
39	Plastics and Articles Thereo	3,248	$148.4M	$1	$1,661	$45,701	$2,646,826	+8.1	+103.5
21	Miscellaneous Edible Prepara	1,154	$136.4M	$1	$562	$118,177	$11,375,910	+10.1	+125.2
18	Cocoa and Cocoa Preparations	468	$134.9M	$1	$700	$288,256	$14,037,695	+7.2	+63.0
48	Paper and Paperboard	1,320	$124.1M	$1	$7,466	$94,045	$3,687,699	+6.9	+59.7
29	Organic Chemicals	2,107	$115.6M	$1	$1,920	$54,849	$5,896,913	+10.9	+131.2
05	Other Animal Products	184	$110.2M	$9	$244,212	$598,995	$4,859,954	+2.4	+6.6
44	Wood and Articles of Wood	1,809	$101.9M	$1	$10,365	$56,333	$1,847,861	+5.2	+47.5
68	Stone, Plaster and Cement Ar	1,439	$90.2M	$1	$7,557	$62,704	$1,480,919	+4.2	+22.2

HS2	Chapter	n	Total (CAD $M)	Min	Median	Mean	Max	Skew	Kurt
98	Special Classification Provi	182	$86.4M	$1	$350	$474,937	$11,017,181	+4.0	+19.3
73	Articles of Iron and Steel	4,063	$82.5M	$1	$731	$20,302	$9,882,227	+31.9	+1248.6

7. Value Distribution — Charts

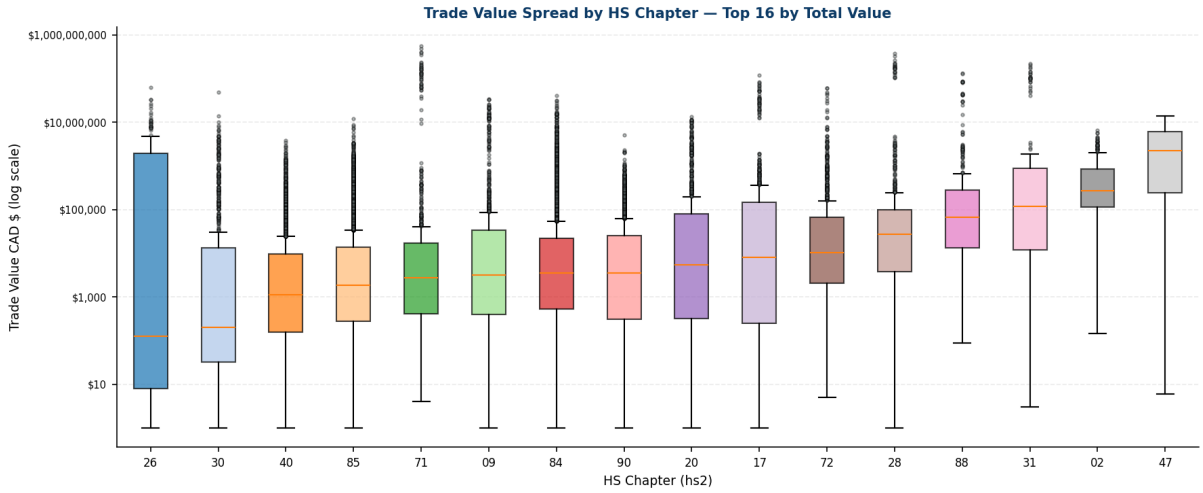
7a. Value Distribution — Log Scale

Log scale is used because trade value spans several orders of magnitude. The overall distribution is right-skewed (positive skew) — a small number of high-value transactions dominate. Top 0.1% clipped for readability.



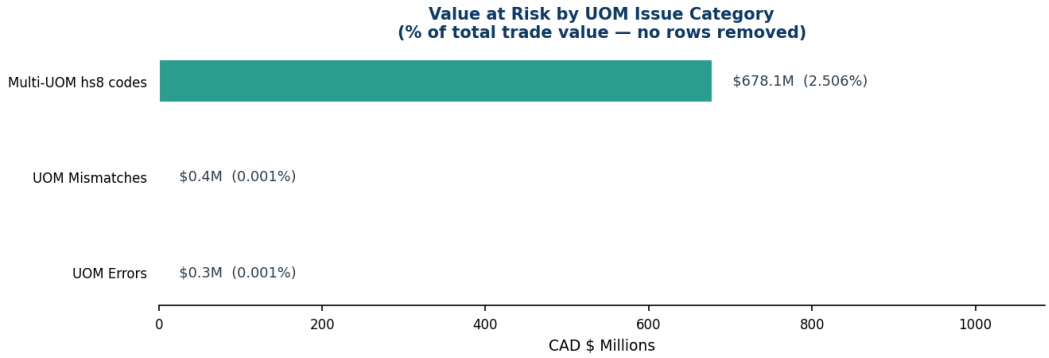
7b. Value Spread by HS Chapter — Boxplot

Each box shows the IQR of transaction values within a chapter (log scale). Circles are statistical outliers. Wide boxes and many outlier points indicate chapters with high value variability — important for model selection.



8. Financial Impact Panel

Value at risk per data quality issue category. No rows were removed — these flags let analysts exclude specific subsets depending on the analysis context.



8a. Top 10 HS8 Codes — UOM Errors by Value

HS8	Chapter	Rows	Value (CAD $)
12099110	Oil Seeds and Oleaginous Fruits	49	$27,094
69022000	Ceramic Products	5	$23,338
71129900	Precious Stones, Metals and Jewelry	1	$22,528
10063000	Cereals	30	$14,801
87032400	Motor Vehicles and Parts	1	$13,720
68029300	Stone, Plaster and Cement Articles	20	$9,963
17011400	Sugars and Sugar Confectionery	4	$9,369
84523000	Nuclear Reactors and Machinery	45	$9,073
11031300	Milling Products, Malt and Starches	36	$8,719
25010010	Salt, Sulfur, Stone and Cement	8	$8,267

8b. Top 10 HS8 Codes — Value Outliers

What the flag means. is_outlier flags transactions whose value falls outside the Tukey inner fence (Q1 – 1.5×IQR to Q3 + 1.5×IQR) for that hs8 code. This is the standard statistical threshold for identifying values that are *unusual relative to the typical range for that commodity*. The flag is purely informational — **no rows are removed**.

Why the flagged value is high (23.2% of total). 9,373 rows are flagged across 1,745 hs8 codes. The high dollar percentage is not a sign of data quality problems — it reflects the structure of Canada-Brazil trade, which is dominated by a small number of commodities with inherently high price variance (precious metals, bulk agricultural contracts, aircraft). The single largest contributor — hs8 71081200 (Precious Stones, Metals and Jewelry) — accounts alone for $2,211,609,986 (35% of all flagged value). Excluding just that one hs8, the remaining flagged value is $4,055,131,255 (15.0% of total trade), which is within a normal range for a dataset of this size.

Why the median is the right model input, not the mean. Because the distribution is right-skewed and flagged transactions are real (not errors), all aggregations and future forecasts should use the **median** as the central tendency measure. The median is robust to extreme values — a $34M coffee shipment does not distort the median the way it distorts the mean. The column median_value_hs8 in the dataset provides this baseline per hs8 code, ready to use.

Recommended action. Review the table below before modelling. For each flagged hs8, decide: (1) include as-is if the commodity is naturally volatile; (2) cap at the upper bound if you need a conservative estimate; (3) exclude if the row is confirmed as a data error. Do not apply a blanket filter — assess per commodity.

HS8	Chapter	Rows	Value (CAD $)
71081200	Precious Stones, Metals and Jewelry	5	$2,211,609,986
09011100	Coffee, Tea, Mate and Spices	48	$748,595,495
28182000	Inorganic Chemicals	1	$373,371,415
84292000	Nuclear Reactors and Machinery	11	$211,190,127
17011400	Sugars and Sugar Confectionery	1	$119,335,340
30049000	Pharmaceutical Products	35	$109,355,255

84295100	Nuclear Reactors and Machinery	16	$83,514,580
84339000	Nuclear Reactors and Machinery	33	$68,790,021
26030000	Ores, Slag and Ash	1	$62,790,671
30043100	Pharmaceutical Products	1	$48,629,167

9. Issue Resolution Guide

A clean dataset does not mean a dataset without open questions — it means every issue has been identified, assessed, and documented. The table below is generated from the current dataset. No rows were removed; all flags are available in the final CSV.

Issue	Scope	Impact	Recommendation
UOM Mismatch — top case: 26030000	26030000 (Chapter: Ores, Slag and Ash)	$121,464,756 — largest UOM mismatch. UOM found: GRM | KGM; expected: KGM.	VALUE is correct and unaffected by UOM discrepancy. Use value only for this hs8. Flag for volume analysis if needed.
Blank UOM — highest-value chapter	Nuclear Reactors and Machinery	$638,710,140 — quantity not published by StatsCan for this commodity type (structural, not a data error).	Use value only. has_quantity=False already flags these rows. No action required.
Multi-UOM within same hs8	215 hs8 codes affected	$678,096,943 — same hs8 code appears with different units across transactions.	Aggregate by value at hs4 level for these codes to avoid unit mixing. uom_status=Mismatch identifies affected rows.
Value outliers flagged (IQR x1.5 per hs8)	All hs8 codes with n >= 4 (3783 total)	9,373 rows flagged ($6,266,741,241 | 23.2% of total value). High value % is driven by a small number of commodities with extreme natural price variance — NOT by data errors (see Section 8b for breakdown).	Review is_outlier=True rows before modelling. Do not remove automatically — most are legitimate large transactions. The flag uses the Tukey inner fence and is informational only.
UOM Errors — qty=0 with physical UOM expected	2,247 rows	$322,460 (0.001% of total). Physical UOM expected by CBSA but quantity is zero.	Consistent with StatsCan suppression for low-volume months. Use value only. No removal required.

9a. UOM Mismatch — Top Cases and Treatment Options

HS8	Chapter	Expected	Found	Rows	Value (CAD $)
26030000	Ores, Slag and Ash	KGM	GRM | KGM	7	$121,464,756
30049000	Pharmaceutical Product	KGM	BLANK | KGM	149	$109,773,849
05119900	Other Animal Products	KGM	KGM | NMB	73	$73,588,835
72249000	Iron and Steel	KGM	KGM | TNE	21	$37,945,854
85176200	Electrical Machinery a	-	BLANK | NMB	159	$33,941,913
90328900	Optical, Medical and P	-	BLANK | NMB	227	$17,782,973
68052000	Stone, Plaster and Cem	KGM	BLANK | KGM	114	$17,332,122
84807100	Nuclear Reactors and M	NMB	BLANK | NMB	30	$17,273,696
84194000	Nuclear Reactors and M	NMB	BLANK | NMB	2	$15,063,164
85269100	Electrical Machinery a	NMB	BLANK | NMB	70	$14,874,646
96091000	Miscellaneous Manufact	GRO	BLANK | GRO	103	$11,655,337
82075000	Tools and Cutlery	NMB	BLANK | NMB	270	$11,300,512
90308400	Optical, Medical and P	NMB	BLANK | NMB	33	$11,262,906
84715000	Nuclear Reactors and M	NMB	BLANK | NMB	46	$10,921,106
84742000	Nuclear Reactors and M	NMB	BLANK | NMB	21	$9,454,700
84212300	Nuclear Reactors and M	NMB	BLANK | NMB	341	$9,039,954
44092990	Wood and Articles of W	MTQ	MTK | MTQ | MTR	132	$7,075,034
85364900	Electrical Machinery a	NMB	BLANK | NMB	197	$7,061,126
44091000	Wood and Articles of W	MTR	MTK | MTQ | MTR	118	$7,047,066
73202000	Articles of Iron and S	KGM	BLANK | KGM	263	$6,900,543

Option	Action	Risk	Recommendation
A — Exclude	Drop Mismatch rows entirely	Value loss; biased totals	Not recommended
B — Impute	Replace quantity with modal UOM conversion	Price distortion	Conditional — only if volume analysis required
C — Flag (current)	Keep all; exclude from UOM-sensitive KPIs	None — safest	Recommended for value-based analysis

Tip: uom_status in dataset_clean.csv lets analysts apply any option without rerunning the pipeline.

10. Data Dictionary — dataset_clean.csv (28 columns)

Authoritative reference for every column in the exported dataset. Source column indicates origin: Raw — StatsCan (field present in source files, text-cleaned only); Derived (computed or extracted during pipeline); Joined — <file> (joined from a reference file); Flag (boolean column computed from data conditions).

Column	Type	Source	Description
period_m	string	Derived	Trade period in YYYY-MM format. Parsed from the raw Period field (YYYY-MM-DD) and truncated to month.
year	int	Derived	Calendar year extracted from period_m. Added to simplify annual grouping.
month	int	Derived	Calendar month (1–12) extracted from period_m. Added to simplify seasonal analysis.
quarter	int	Derived	Calendar quarter (1–4) extracted from period_m. Added to simplify quarterly grouping.
flow	string	Derived	Trade direction: Import or Export. Added during ingestion to tag which source file each row came from.
province	string	Raw — StatsCan	Canadian province or territory. Text-cleaned: Unicode normalised, non-ASCII stripped, whitespace collapsed.
country	string	Raw — StatsCan	Partner country name. Text-cleaned same as province. Always Brazil in this dataset.
description	string	Derived	Product description extracted from the raw Commodity field (text after the first ' - ' separator). Non-ASCII characters removed; leading/trailing/double spaces collapsed.
hs2	string	Derived	HS Chapter code — first 2 digits of the raw Commodity code. Zero-padded string. Highest-level product grouping.
hs4	string	Derived	HS Heading code — first 4 digits of the raw Commodity code. Zero-padded string.
hs6	string	Derived	HS Subheading code — first 6 digits. International standard level. Zero-padded string.
hs8	string	Derived	HS Tariff item — first 8 digits. Export-level granularity. Used as the common join key between Import and Export flows. Zero-padded string.
hs10	string	Derived	HS Statistical suffix — 10 digits. Import-level only (H10 source file). NaN for all Export rows — the export file (H8) does not contain the 10-digit suffix.
section	string	Joined — hs_mapping.xlsx	HS Section code (Roman numeral, e.g. VII). Joined from hs_mapping.xlsx on hs2.
section_name	string	Joined — hs_mapping.xlsx	HS Section name (e.g. Plastics and Rubber). Joined from hs_mapping.xlsx on hs2.
chapter_name	string	Joined — hs_mapping.xlsx	HS Chapter name (e.g. Rubber and Articles Thereof). Joined from hs_mapping.xlsx on hs2.
value	int	Raw — StatsCan	Trade value in Canadian dollars (CAD). Raw column: Value ($). Coerced to int; NaN → 0. Zero values indicate StatsCan confidentiality suppression — see is_suppressed. Primary metric for all analyses in this report.
quantity	int	Raw — StatsCan	Physical quantity traded. Raw column: Quantity. Coerced to int; NaN → 0. Zero when uom_text is Blank (StatsCan does not publish quantity for all commodity types) or below rounding threshold. NOT used as a primary analysis metric — see has_quantity.
uom_text	string	Raw — StatsCan	Unit of measure as published by StatsCan. Raw column: Unit of measure. Text-cleaned (same as province). Values: Weight in kilograms, Number, Blank, etc.
uom_code	string	Derived	Standardised UOM code mapped from uom_text via UOM_TEXT_TO_CODE lookup table (e.g. KGM, NMB, BLANK). UNKNOWN if uom_text not in the lookup.
uom_status	string	Derived	UOM audit result against CBSA tariff reference (t2026_uom.xlsx): OK = passes audit | Error = physical UOM expected but quantity is zero | Mismatch = UOM used differs from CBSA expectation.
has_quantity	bool	Flag	True when uom_text != Blank. Derived flag. Rows with has_quantity=False must be excluded from any volume-based analysis.
is_ch98	bool	Flag	True when hs2 == 98 (Canadian Special Classification Provisions). Derived flag. Chapter 98 covers non-commercial border movements: repairs sent abroad, settlers effects, traveller goods, low-value postal shipments, charitable donations, and returning Canadian goods. Must be excluded from commodity trend analysis and forecast models.
is_suppressed	bool	Flag	True when value == 0. Derived flag. StatsCan withholds value and quantity when a single company dominates a trade cell in a given month (Disclosure Control Policy). Must be excluded from all value aggregations.

Column	Type	Source	Description
is_outlier	bool	Flag	True when value falls outside the IQR x1.5 bounds for the same hs8 code (minimum 4 observations required). Uses the Tukey inner fence method on trade VALUE. Outliers are FLAGGED, never removed — they represent genuine large transactions in naturally volatile commodities (bulk agricultural, precious metals, aircraft). Analysts decide per hs8 whether to include or exclude them for modelling.
median_value_hs8	float	Derived	Median trade value across all non-suppressed transactions sharing the same hs8 code (min 4 observations). Baseline for IQR x1.5 outlier detection. NaN when hs8 has fewer than 4 observations.
lower_bound_hs8	float	Derived	Lower IQR bound = max(Q1 - 1.5×IQR, 0). Clipped at 0 — trade value cannot be negative. NaN when hs8 has fewer than 4 observations.
upper_bound_hs8	float	Derived	Upper IQR bound = Q3 + 1.5×IQR. Transactions above this threshold are flagged as is_outlier=True. NaN when hs8 has fewer than 4 observations.